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1TPIFI-2
18/03/2026

L5 - DataTransfer

1. Introduction

This document explains the functionality and structure of a PHP script designed to receive sensor data via HTTP POST and store it in a MySQL database. The script is intended for use in a Portable Indoor Feedback system.

2. Source Code

```
<?php
/**
 * Data Transfer Script for Portable Indoor Feedback
 * Purpose: Receives sensor data via HTTP POST and stores it in the central
database.
 */

// Database connection parameters
$servername = "localhost";
$username   = "db_user";
$password   = "db_password";
$dbname     = "feedback_db";

// 1. Establish connection to the database server
$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// 2. Check if data is being sent via HTTP POST
if ($_SERVER["REQUEST_METHOD"] == "POST") {

    // Collect and sanitize input data
    $serial_number = mysqli_real_escape_string($conn, $_POST['serial_number']);
    $timestamp     = mysqli_real_escape_string($conn, $_POST['timestamp']);
    $measure_val   = mysqli_real_escape_string($conn, $_POST['value']); //
Example measurement

    // 3. Prepare the SQL statement to ensure all values are stored correctly
    $sql = "INSERT INTO measurements (serial_number, timestamp, measured_value)
            VALUES ('$serial_number', '$timestamp', '$measure_val')";

    if ($conn->query($sql) === TRUE) {
        echo "Data transferred and stored successfully.";
    } else {
        echo "Error: " . $sql . "<br>" . $conn->error;
    }
}

// 4. Close the connection
$conn->close(); ?>
```

3. Code Explanation

3.1 Database Configuration

The script begins by defining database connection parameters including server name, username, password, and database name. These values are required to establish a connection to the MySQL database.

3.2 Database Connection

A new MySQLi connection is created using the provided credentials. If the connection fails, the script stops execution and outputs an error message.

3.3 HTTP POST Check

The script verifies that the incoming request uses the HTTP POST method. This ensures that data is being securely sent from the client (sensor or form) to the server.

3.4 Data Collection and Sanitization

Input data such as serial number, timestamp, and measurement value are retrieved from the POST request. The mysqli_real_escape_string() function is used to sanitize inputs and prevent SQL injection attacks.

3.5 SQL Query Execution

An SQL INSERT statement is constructed to store the received data into the "measurements" table. If the query executes successfully, a success message is displayed; otherwise, an error message is shown.

3.6 Connection Closure

After processing the request, the database connection is closed to free up server resources.

4. System Overview

This script acts as a bridge between sensor devices and the central database. It receives, processes, and stores environmental or feedback data for later analysis.

5. Data Flow

1. Sensor sends data via HTTP POST.
2. Script receives and validates the request.
3. Data is sanitized and inserted into the database.
4. Confirmation or error message is returned.

6. Troubleshooting

Connection issues: Verify database credentials and server status.
Data issues: Ensure POST parameters match expected names (serial_number, timestamp,

value).
Permission issues: Ensure the script has access to the database.